# Rahul Yedida

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## Education

**North Carolina State University** — Raleigh, NC

*Ph.D. Computer Science* — *Aug 2019 - Jul 2024*

- Advisor: *Dr. Tim Menzies*
- Dissertation: *Guidelines for the Application of Neural Technologies in Software Analytics (or: How to Do More with Less in SE)*

**PES University** — Bangalore, India

*B.E. Computer Science – Advisor: Dr. Snehanshu Saha* — *Aug 2015 - May 2019*

## Employment

**LexisNexis Legal & Professional** — Raleigh, NC

*Senior Data Scientist II* — *Oct 2026 - Present*

- Fine-tuned small structured-decision models to match LLM-baseline performance (macro-F1 92.7%) at 177 ms median latency on-device, after benchmarking 15 models across 6 tasks.

*Senior Data Scientist I* — *May 2024 - Oct 2026*

- Technical lead for an agentic legal drafting system; led its migration from a RAG pipeline and own its ongoing development.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Built a contrastive-learning filter using InfoNCE with alignment and uniformity losses to remove unrelated legal passages from search results; median filtering latency of 12 ms across 40 queries.
- Built team developer tooling: custom Python lint rules beyond `ruff`'s set, enforced in CI; a Rust MITM proxy with request replay that teammates use to debug without waiting on upstream responses; and a web UI for visualizing intermediate agent steps.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

**North Carolina State University** — Raleigh, NC

*PhD Student* — *Aug 2019 - Jul 2024*

- **State-of-the-art hyper-parameter optimization:** Derived loss-curvature estimates for smoothness- and flatness-guided search; the smoothness-based method matched or beat prior SOTA on 17 of 19 software datasets.
- **Better, faster deep learning for SE:** Introduced fuzzy oversampling and optimized neural-network hyper-parameters, surpassing prior SOTA with median relative gains of 15% in defect-prediction F1 and 12% in AUC across four code smells. Built tuned feedforward models that matched or beat prior work across 15 issue-lifetime benchmarks in minutes instead of hours.
- **Microservice partitioning:** Developed a graph neural network partitioner with a custom weighted loss; reduced the share of cross-service calls by a median 66% versus prior SOTA across four applications.
- **Semi-supervised learning:** Combined oversampling, nearest-neighbor labeling, and hyper-parameter optimization for state-of-the-art static-analysis warning triage: median AUC 0.92 on eight Java projects using 10% of training labels.
- **Teaching:** TA for 830 students over five semesters in C programming and software engineering.

## Recent Publications

See full list on Google Scholar.

1. Nagarsekar, A., Bhat, M. P. A., Nesarkar, A., Godhwani, V., **Yedida, R.**, Challa, A., Danda, S., & Saha, S. (2026). A Generalisation Signal Need Not Be a Model-Selection Signal. In *I Can't Believe It's Not Better: Failure Modes of AI in Biology Workshop (NeurIPS 2026)*.

2. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering, 51(6)*.

3. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM, 66(2), 75-81*.

4. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering, 49(4), 2856-2872*.

5. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications, 219, 119673*.

6. **Yedida, R.**, & Menzies, T. (2022). How to Improve Deep Learning for Software Analytics (a case study with code smell detection). In *2022 IEEE/ACM 19th International Conference on Mining Software Repositories (MSR)*.

7. **Yedida, R.**, & Menzies, T. (2021). On the Value of Oversampling for Deep Learning in Software Defect Prediction. *IEEE Transactions on Software Engineering, 48(8), 3103-3116*.

8. Agrawal, A., Yang, X., Agrawal, R., **Yedida, R.**, Shen, X., & Menzies, T. (2021). Simpler Hyperparameter Optimization for Software Analytics: Why, How, When?. *IEEE Transactions on Software Engineering, 48(8), 2939-2954*.

9. Yang, X., Chen, J., **Yedida, R.**, Yu, Z., & Menzies, T. (2021). Learning to recognize actionable static code warnings (is intrinsically easy). *Empirical Software Engineering, 26(3), 1-24*.

10. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2021). Lessons learned from hyper-parameter tuning for microservice candidate identification. *Proceedings of the thirty-sixth IEEE/ACM International Conference on Automated Software Engineering (ASE)*.

11. **Yedida, R.**, & Saha, S. (2021). Beginning with Machine Learning: A Comprehensive Primer. *The European Physical Journal Special Topics*, 230(10), 2363-2444.

12. Saha, S., Nagaraj, N., Mathur, A., **Yedida, R.**, & Sneha, H. R. (2020). Evolution of novel activation functions in neural network training for astronomy data: habitability classification of exoplanets. *The European Physical Journal Special Topics, 229(16), 2629-2738*.

13. **Yedida, R.**, Saha, S., & Prashanth, T. (2020). LipschitzLR: Using theoretically computed adaptive learning rates for fast convergence. *Applied Intelligence, 1-19*.

14. Sridhar, S., Saha, S., Shaikh, A., **Yedida, R.**, & Saha, S. (2020, July). Parsimonious Computing: A Minority Training Regime for Effective Prediction in Large Microarray Expression Data Sets. In *2020 International Joint Conference on Neural Networks (IJCNN) (pp. 1-8)*.

15. Khaidem, L., **Yedida, R.**, & Theophilus, A. J. (2019, November). Optimizing Inter-nationality of Journals: A Classical Gradient Approach Revisited via Swarm Intelligence. In *International Conference on Modeling, Machine Learning and Astronomy (pp. 3-14). Springer, Singapore*.

## Preprints

1. Hearn, T., Evans, W., **Yedida, R.**, Abdelshiheed, M., Rebei, A., & Khazaeli, S. (2025). Automated Legal Brief Drafting via Iterative Retrieval-Augmented Generation.

2. Rashid, P., Pal, S., Samudrala, V. K., **Yedida, R.**, & Gehringer, E. (2025). Enhancing Peer Assessment Reliability: A Probabilistic Approach Integrating Formative and Summative Feedback.

3. Mehendale, S., Challa, A., **Yedida, R.**, Danda, S., Sarkar, S., & Saha, S. (2025). Radon–Nikodym Derivative: Re-imagining Anomaly Detection from a Measure Theoretic Perspective.

4. **Yedida, R.**, & Saha, S. (2024). Strong convexity-guided hyper-parameter optimization for flatter losses.

## Funding

**$5,000**, Google Cloud Academic Research Grant, Feb 2022

## Service to Profession

**Contributing Member**, Python Software Foundation (Mar 2021 - Present)

**Guest Editor**, IEEE Software SI: The Impact of AI on Productivity and Code 2025; ASEj SI: Replications and Negative Results (RENE) 2025; EMSE SI: Replications and Negative Results (RENE) 2025

**Workshop Co-Chair**, Intelligent Software Engineering (ASE '25, ICST '26); Replications and Negative Results (ASE '24)

**PC Member**, AAAI 2025-2027; FSE 2026-2027; ICSE 2026-2027; EDM 2026; ASE Industry Track 2026; AI Foundation Models and Software Engineering (FORGE) 2024; ICSME Artifact Evaluation Track, 2021-2023; ASE Artifact Evaluation Track, 2022; International Conference on Modeling, Machine Learning, and Astronomy (MMLA), 2019

**Reviewer**, TOSEM 2026; TSE 2025-2026; NeurIPS 2023, 2025; EMSE 2021, 2025; ASEj 2025; ICML 2024-2025; ICLR 2024-2025; NCAA 2023-2025; TMLR 2024; Neural Processing Letters 2023-2024; Artificial Intelligence Review 2023; J. Big Data, 2023; ASE 2023; IEEE SSCI 2020

## Honors

**Distinguished Reviewer, ICSE 2026** Mar 2026

**Google Developer Expert - ML and GCP** Mar 2025 - Present

A global network of experts recognized by Google for deep technical expertise, community leadership, and mentorship.

**Google Cloud Champion Innovator - Cloud AI/ML** Oct 2022 - Mar 2025

A Googler-nominated global network of 500+ technical experts in Google Cloud; merged into the Google Developer Experts program in 2025.

**Google Cloud Research Innovator** Feb 2022

Research Innovators are a global community of researchers driving scientific breakthroughs with Google Cloud.

## Invited Talks

**Feb 2024**, "*Improving deep learning performance using theoretical ML*" at BITS Pilani, KK Birla Goa Campus, India

## Skills

**Languages**: Rust, Python, TypeScript, Java, C++, Gleam

**ML/AI**: PyTorch, Keras, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK

**Frameworks**: React, FastAPI, Litestar, FastAPI

**Databases**: LanceDB, SQL, MongoDB, DynamoDB

**Cloud**: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)